Extracting the 0^{++} Glueball Mass on the Lattice: from the Euclidean Path Integral to a Learned Variational Operator

Francesco Passante

August 2, 2026

Abstract

Glueballs are bound states of pure gauge fields, and their masses are among the most natural — and notoriously noisy — predictions of lattice gauge theory. This note develops, from first principles, everything needed to extract the lightest scalar (0^{++}) glueball mass in $SU(2)$ lattice gauge theory: the Euclidean path integral and the transfer matrix, the spectral decomposition of connected correlators, the effective mass and its variational bound, the signal-to-noise catastrophe specific to glueballs, and the three classical tools that tame it — link smearing, the variational method (GEVP), and the anisotropic lattice. We then document an actual extraction on an anisotropic $SU(2)$ ensemble ($12^3 \times 24$, $\beta = 2.4$, $\xi = 3$), where a multi-level smearing GEVP resolves a clean plateau at $m\,a_t \approx 0.33$. Finally, we replace the hand-built operator basis by a gauge-equivariant neural network (GELT) trained on a Rayleigh-quotient loss whose converged value *is* the mass. Fed multi-level smeared input channels and constrained to a single timeslice (so the transfer-matrix bound applies), the learned operator saturates the variational bound at the classical anchor, plateaus from $\Delta = 1$, and carries significantly less excited-state contamination than the classical GEVP (3.9σ from a correlated-difference jackknife) while agreeing with it on the mass itself. A from-scratch replication on an independently sampled ensemble reproduces the result; the two ensembles combine to a 3.6σ excess in ground-state overlap. A closing exploratory section reads the learned operator's gauge-invariant attention weights out as a measurement, finding its advantage concentrated in a single local attention head.

Contents

1 Introduction

In a pure Yang–Mills theory — QCD with the quarks removed — the only physical states are *glueballs*: color-singlet bound states of gluons, held together by the same self-interaction that makes the theory confining. Their spectrum is a parameter-free prediction of the gauge dynamics, and computing it was one of the first triumphs of lattice gauge theory [2, 12]. The lightest state carries the quantum numbers of the vacuum, $J^{PC} = 0^{++}$, and it is the target of everything that follows.

Extracting a glueball mass is conceptually simple — measure a Euclidean correlator, read off its exponential decay — and practically hard: the signal decays exponentially while the noise does not, so the mass must be read at very small time separations, which is only possible if the measuring operator overlaps strongly onto the ground state. The entire craft of glueball spectroscopy is the construction of such operators.

This note has two halves. Sections 2–7 build the standard theory and practice from the ground up, assuming only basic familiarity with lattice gauge theory, and document a classical measurement of the $SU(2)$ 0^{++} mass on an anisotropic lattice. Sections 8–9 then reformulate the operator construction as a machine-learning problem: since the optimal operator is the maximizer of a Rayleigh quotient of the transfer matrix, a gauge-equivariant network trained to minimize $-C(1)/C(0)$ *is* a variational glueball operator, and its converged loss *is* (minus the exponential of) the mass. The final result is a learned operator that variationally subsumes the hand-built smearing basis it was validated against.

2 From the path integral to the lattice

2.1 The Euclidean path integral

Continuum expectation values of a quantum field theory can be written, after Wick rotation $t \to -i\tau$, as statistical averages with a real, positive weight,

$$\langle O \rangle \;=\; \frac{1}{Z} \int \mathcal{D}A \; O[A] \, e^{-S_E[A]}, \qquad Z = \int \mathcal{D}A \; e^{-S_E[A]}, \tag{1}$$

where S_E is the Euclidean action. Two things make this formulation the foundation of lattice gauge theory. First, e^{-S_E} is a probability weight (up to normalization), so (1) can be estimated by Monte Carlo importance sampling. Second — and this is the part that matters for spectroscopy — Euclidean time evolution is generated by $e^{-\tau \hat{H}}$ rather than $e^{-it\hat{H}}$: excited states are *exponentially suppressed* rather than oscillating, so the large-τ behaviour of correlation functions isolates the lowest-energy state in a given channel. Masses are read off exponential decays. Section 3 makes this precise via the transfer matrix.

2.2 Gauge fields on a lattice: links, plaquettes, Wilson action

Wilson's discretization [1] replaces spacetime by a hypercubic lattice Λ with spacing a, and the gauge potential $A_\mu(x)$ by group-valued *link variables*

$$U_\mu(x) \;=\; \mathcal{P} \exp\Big(ig \int_x^{x+a\hat{\mu}} A \cdot dx \Big) \;\in\; SU(N), \tag{2}$$

the parallel transporters along the elementary edges of the lattice. A gauge transformation is a group element $\Omega(x)$ at every site, acting as

$$U_\mu(x) \;\longrightarrow\; \Omega(x) \, U_\mu(x) \, \Omega^\dagger(x + a\hat{\mu}). \tag{3}$$

Any product of links along a *closed* loop therefore transforms by conjugation at its base point, and its trace is exactly gauge invariant. The smallest such loop is the *plaquette*,

$$P_{\mu\nu}(x) \;=\; U_\mu(x) \, U_\nu(x + a\hat{\mu}) \, U_\mu^\dagger(x + a\hat{\nu}) \, U_\nu^\dagger(x), \tag{4}$$

and the Wilson action is the sum of plaquettes,

$$S[U] \;=\; \beta \sum_{x,\, \mu<\nu} \Big(1 - \tfrac{1}{N} \, \mathrm{Re} \, \mathrm{Tr} \, P_{\mu\nu}(x) \Big), \qquad \beta = \frac{2N}{g^2}, \tag{5}$$

which reproduces $\frac{1}{4} F^a_{\mu\nu} F^a_{\mu\nu}$ in the classical continuum limit $a \to 0$. The bare coupling β is the only parameter; the lattice spacing $a(\beta)$ is not an input but an output, set by the dynamics (asymptotic freedom drives $a \to 0$ as $\beta \to \infty$). All results below are quoted in lattice units, i.e. as dimensionless products such as $m\,a$.

2.3 Expectation values by Monte Carlo

The lattice path integral is an ordinary (huge-dimensional) integral over the compact group manifold, with Haar measure on every link:

$$\langle O \rangle \;=\; \frac{1}{Z} \int \prod_{x,\mu} dU_\mu(x) \; O[U] \, e^{-S[U]}. \tag{6}$$

A Markov-chain Monte Carlo algorithm generates an ensemble of configurations $\{U^{(1)}, \ldots, U^{(N)}\}$ distributed as e^{-S}, and $\langle O \rangle \approx \frac{1}{N} \sum_i O[U^{(i)}]$ with an error $\propto 1/\sqrt{N_{\text{eff}}}$, where N_{eff} accounts for autocorrelation along the chain (Section 6). Everything in this note runs in the gauge group $SU(2)$, which is cheap, confining, and admits exact single-link updates.

3 Masses from Euclidean correlators

3.1 The transfer matrix

The Euclidean lattice theory on a lattice with N_t timeslices and periodic temporal boundary conditions is equivalent to a quantum system with Hamiltonian $\hat{H}$ acting on a Hilbert space of spatial-field states: the partition function factorizes as

$$Z \;=\; \mathrm{Tr}\,\hat{T}^{\,N_t}, \qquad \hat{T} \;=\; e^{-a_t\hat{H}}, \tag{7}$$

where $\hat{T}$, the *transfer matrix*, propagates the system by one timeslice (a_t is the temporal lattice spacing). For the Wilson action $\hat{T}$ is self-adjoint and positive (*reflection positivity* [3]), which is what makes everything below a theorem rather than a heuristic: $\hat{T}$ has a complete set of eigenstates $|n\rangle$ with real eigenvalues $e^{-a_t E_n}$, $E_0 \le E_1 \le \dots$, and we measure energies relative to the vacuum, $E_0 = 0$.

An operator $O(\vec{x}, t)$ built from the *spatial* links of timeslice t only corresponds to a Schrödinger-picture operator $\hat{O}$ on this Hilbert space. This locality-in-time condition looks like a technicality; it will return twice with teeth — once for smearing (Section 5.1) and once, decisively, for the neural network (Section 8.3).

3.2 Spectral decomposition of the correlator

Insert complete sets of transfer-matrix eigenstates into a two-point function at temporal separation Δ (in units of a_t):

$$\langle O(t{+}\Delta)\, O(t)\rangle \;=\; \frac{1}{Z}\,\mathrm{Tr}\big[\hat{T}^{\,N_t-\Delta}\,\hat{O}\,\hat{T}^{\,\Delta}\,\hat{O}\big] \;\xrightarrow[N_t\gg\Delta]{}\; \sum_n \big|\langle 0|\hat{O}|n\rangle\big|^2\, e^{-E_n\Delta\, a_t}. \tag{8}$$

Every term is positive, and the decay rates are the energies of the states the operator couples to. At large Δ the lightest such state dominates: its energy is read off the exponential tail. (On a finite periodic lattice the exponential is replaced by $e^{-E_n\Delta a_t} + e^{-E_n(N_t-\Delta)a_t} \propto \cosh$, symmetric about $\Delta = N_t/2$; this matters only near the midpoint.)

3.3 Zero momentum, vacuum subtraction, effective mass

Three standard refinements turn (8) into a mass measurement.

Zero-momentum projection. Summing the operator over a spatial slice,

$$\bar{O}(t) \;=\; \sum_{\vec{x}} O(\vec{x}, t), \tag{9}$$

projects onto momentum $\vec{p} = 0$, so the energies in the decomposition are *masses*, $E_n = m_n$, with no kinetic contamination. It also averages over L^3 sites, improving statistics.

Vacuum subtraction. The 0^{++} channel shares the quantum numbers of the vacuum, so $\langle\bar{O}\rangle \neq 0$ and the $n = 0$ term of (8) is a Δ-independent constant $|\langle 0|\hat{O}|0\rangle|^2$ that would bury the signal. One therefore measures the *connected* correlator,

$$C(\Delta) \;=\; \langle \bar{O}(t_0{+}\Delta)\,\bar{O}(t_0)\rangle - \langle\bar{O}\rangle^2 \;=\; \sum_{n>0} \big|\langle 0|\hat{\bar{O}}|n\rangle\big|^2\, e^{-m_n\Delta\, a_t}, \tag{10}$$

averaged over all time origins t_0 (time-translation invariance buys another factor N_t of statistics). A structural gift of the 0^{++} channel: since it *is* the vacuum channel, after subtraction the lightest surviving state is automatically the scalar glueball — no quantum-number projection (spin, parity, charge conjugation on the cubic group) is needed for the ground state.

Effective mass. Define

$$m_{\text{eff}}(\Delta) \;=\; \log \frac{C(\Delta)}{C(\Delta+1)}. \tag{11}$$

Because every term in (10) is positive, $m_{\text{eff}}(\Delta) \geq m_G$ for all Δ and $m_{\text{eff}}(\Delta) \to m_G\, a_t$ monotonically from above as $\Delta \to \infty$: the correlator is a *variational upper bound* machine. In practice one looks for the *plateau* — the range of Δ where m_{eff} has stopped descending (excited states have decayed) but the error bars have not yet exploded. The whole game is to make that window exist.

4 Why the glueball is hard: signal-to-noise

For most hadrons the plateau window is wide. For glueballs it is a crack in the wall, for a reason first made sharp by Lepage [8]. The Monte Carlo variance of the correlator estimate is itself a correlator,

$$\mathrm{Var}\big[C(\Delta)\big] \;\sim\; \langle \bar{O}^2(t_0{+}\Delta)\, \bar{O}^2(t_0)\rangle - C(\Delta)^2, \tag{12}$$

and the operator $\bar{O}^2$ couples to the *vacuum*: the variance approaches a Δ-independent constant. The signal decays as $e^{-m_G \Delta a_t}$ while the noise does not, so

$$\frac{\text{signal}}{\text{noise}} \;\sim\; \sqrt{N}\; e^{-m_G \Delta\, a_t}. \tag{13}$$

The scalar glueball is *heavy* — $m_G\, a$ is order one on typical lattices — so the signal is gone within two or three timeslices, and by (13) doubling the statistics buys only $\log\sqrt{2}/(m_G a_t)$ extra slices. More configurations are nearly useless; the plateau must be dragged to *small* Δ, where the noise is still small. Since $m_{\text{eff}}(\Delta)$ starts high and descends as excited states die off, dragging the plateau to small Δ means building an operator with large *ground-state overlap*: $|\langle 0|\hat{\bar{O}}|1\rangle|^2$ dominant in (10) already at $\Delta = 1$. Everything in the next section is a technique for manufacturing overlap.

5 The classical toolkit

5.1 Smeared operators

The simplest scalar glueball operator is the spatial plaquette itself, summed over orientations to make a rotational scalar. It is a terrible interpolating operator: a single *thin-link* loop of extent a probes UV fluctuations, not the physical glueball, whose wavefunction has a size of order the confinement scale. Its m_{eff} starts far above m_G and descends too slowly to plateau before the noise wall of (13).

The fix is *link smearing*: replace each spatial link by a gauge-covariant local average of itself and its neighbouring *staples* (the three-link detours connecting the same endpoints). One APE step [7] is

$$U_j(x) \;\longrightarrow\; \mathrm{Proj}_{SU(N)}\Big[(1-\alpha)\, U_j(x) + \frac{\alpha}{n_{\text{st}}} \sum_{\text{spatial staples}} S_j(x)\Big], \tag{14}$$

iterated n times; operators built from smeared links are spatially extended over a radius that grows with n, matching the physical size of the state and suppressing the UV contamination. Two rules are absolute:

- **Smear spatial links only, never in time.** Temporal smearing makes the operator depend on several timeslices at once, destroying the transfer-matrix interpretation of (8) — the "operator on timeslice t" premise — and with it the variational bound.

- **Smearing is not cooling.** Cooling/gradient flow drives the *configuration* toward classical solutions and changes the measured physics; operator smearing changes only the *operator*, leaving the ensemble untouched. Both are built from staples; they answer different questions.

There is no single best smearing level — too little leaves UV contamination, too much washes the operator out — which motivates using *several* levels at once:

5.2 The variational method (GEVP)

Given a basis of $n_{\rm op}$ operators $\bar{O}_i$ (here: the scalar operator at APE levels $0, 2, 4, 6$), measure the full connected correlator *matrix*

$$C_{ij}(\Delta) \;=\; \langle \bar{O}_i(t_0{+}\Delta)\,\bar{O}_j(t_0)\rangle - \langle \bar{O}_i\rangle\langle \bar{O}_j\rangle, \tag{15}$$

and solve the *generalized eigenvalue problem* (GEVP) [9, 10, 11]

$$C(\Delta)\,v_n \;=\; \lambda_n(\Delta, t_0)\;C(t_0)\,v_n, \tag{16}$$

at a fixed reference time t_0 (here $t_0 = 1$). The eigenvector v_n defines the linear combination $\sum_i (v_n)_i \bar{O}_i$ that optimally isolates state n within the span of the basis, and the eigenvalues decay with the *individual* state energies,

$$\lambda_n(\Delta, t_0) \;=\; e^{-m_n(\Delta - t_0)\,a_t}\big(1 + \mathcal{O}(e^{-\delta m\,\Delta\,a_t})\big), \tag{17}$$

with corrections controlled by the gap δm to the first state *outside* the basis span — parametrically smaller than the single-operator contamination. Effective masses are then read from the eigenvalues, $m_{\rm eff}^{(n)}(\Delta) = \log[\lambda_n(\Delta)/\lambda_n(\Delta+1)]$, for $\Delta \geq t_0$. This "multi-level smearing basis + GEVP" construction is the Morningstar–Peardon strategy [12] that underlies every modern glueball spectrum.

One numerical remark from practice: at low statistics the estimated $C(t_0)$ can be indefinite, so a naive Cholesky-based solver fails. The implementation used here whitens with the eigendecomposition of $C(t_0)$ and floors near-zero eigenvalues — adequate for four well-separated smearing levels, though for larger bases the flooring should be replaced by truncation (projecting the noisy directions out rather than blowing them up).

5.3 Anisotropic lattices

Smearing and the GEVP fix the *overlap*; they cannot manufacture timeslices. If $m_G\,a_t \sim 1$, the correlator falls by e^{-1} per slice and even a perfect operator has only 2–3 usable points — too few to demonstrate a plateau. The field's standard cure [12] is the *anisotropic lattice*: a temporal spacing finer than the spatial one, $a_t = a_s/\xi$ with anisotropy $\xi > 1$, so the same physical decay is sampled by ξ times more slices. At tree level this is a reweighting of the Wilson action by plaquette orientation,

$$S \;=\; \beta_s \sum_{\text{spatial } P}\Big(1 - \tfrac{1}{N}\mathrm{Re\,Tr}\,P\Big) + \beta_t \sum_{\text{temporal } P}\Big(1 - \tfrac{1}{N}\mathrm{Re\,Tr}\,P\Big), \qquad \beta_s = \frac{\beta}{\xi}, \quad \beta_t = \beta\,\xi. \tag{18}$$

The masses extracted from temporal decays come out in units of a_t; the conversion $m\,a_s = \xi\,m\,a_t$ uses the *renormalized* anisotropy ξ_R, which quantum corrections shift away from the bare ξ (we measure the shift but do not tune it away; see Section 7.4).

6 Ensemble generation and error analysis

6.1 Heat-bath and overrelaxation for $SU(2)$

Spectroscopy needs many well-decorrelated configurations, and the workhorse Metropolis algorithm decorrelates slowly (each accepted move is a small random step). $SU(2)$ admits two exact, tuning-free local updates that together beat this critical slowing down:

Heat-bath [4, 5]. The action is linear in any single link: $S \supset -\frac{\beta}{N}\mathrm{Re}\,\mathrm{Tr}[U_\mu(x)A^\dagger]$, where A is the sum of the staples around that link (with the $\xi^{\pm 1}$ weights of (18) folded in). For $SU(2)$ a sum of group elements is proportional to a group element, $A = k\,V$ with $V \in SU(2)$, so the single-link conditional distribution is known in closed form and can be sampled *exactly* — the new link is completely independent of the old one.

Overrelaxation [6]. The reflection $U \to V^\dagger U^\dagger V^\dagger$ preserves the action exactly (microcanonical) while moving as far as possible across the constant-action surface — pure decorrelation at negligible cost. The production sweep used below is one heat-bath sweep followed by $n_{\mathrm{OR}} = 4$ overrelaxation sweeps, in a checkerboard (even/odd) pattern so that whole sublattices update in parallel.

6.2 Autocorrelation

Successive configurations of the chain are correlated; the effective sample size is $N_{\mathrm{eff}} = N/(2\tau_{\mathrm{int}})$, where the integrated autocorrelation time τ_{int} of the *measured observable* (estimated with Madras–Sokal automatic windowing [13]) sets the number of sweeps $n_{\mathrm{skip}} \gtrsim 2\tau_{\mathrm{int}}$ to discard between saved configurations. Crucially, τ_{int} must be measured on the smeared glueball operator, not the plaquette — extended observables decorrelate more slowly.

6.3 Jackknife errors

All quoted errors are leave-one-out jackknife errors over configurations: recompute the full analysis (correlator, GEVP, m_{eff}) on N subsamples each omitting one configuration; the spread of the results estimates the error of the full-sample result, propagating correctly through every nonlinearity (ratios, logs, eigenvalues). When residual autocorrelation is suspected, the omitted unit is a contiguous *block* of configurations rather than one ("blocked jackknife"), which keeps the independence assumption honest. Two further uses appear below: neighbouring-Δ points of a jackknifed m_{eff} curve are strongly correlated, so "the error bars overlap" can overstate consistency; and when comparing two estimates measured on the *same* configurations, the jackknife of their *difference* cancels the correlated fluctuations and is the honest significance test (Section 9.2).

7 The classical measurement, run by run

All ensembles are $SU(2)$, sampled with heat-bath + overrelaxation. The operator is the orientation-summed spatial 1×1 Wilson loop (the spatial plaquette scalar), at APE levels $0, 2, 4, 6$; the analysis is the connected correlator (10), m_{eff} (11), and the GEVP (16) with $t_0 = 1$, all errors jackknifed.

7.1 Runs 0–1: the isotropic lattice fails

On an isotropic 12^4 lattice at $\beta = 2.4$ with $N = 2000$ configurations, the single smeared operator shows only a weak $m\,a \approx 0.8$ at $\Delta = 1$–2 and slides into noise (negative m_{eff}, huge bars) by

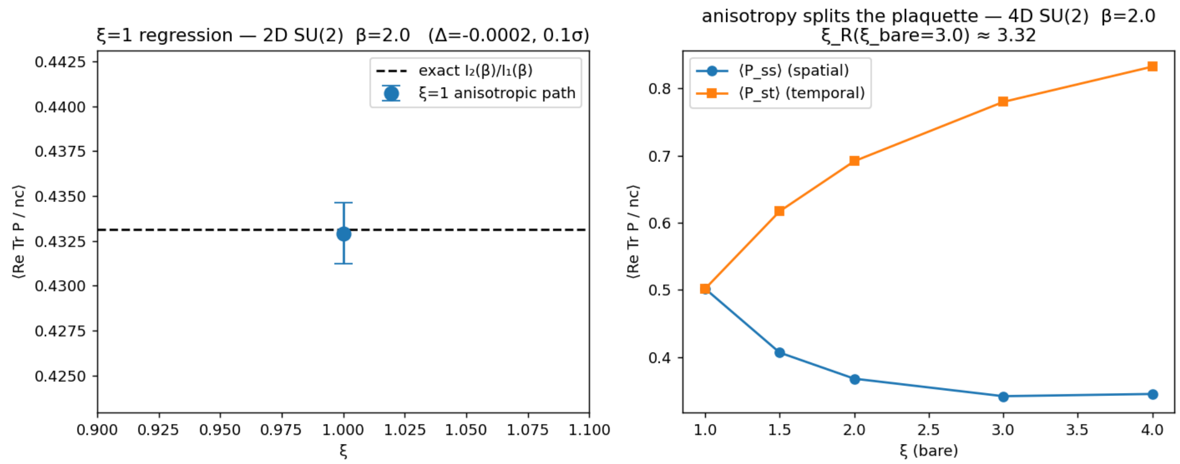

Figure 1: Validation of the anisotropic $SU(2)$ action. Left: the $\xi = 1$ code path reproduces the exact 2D mean plaquette $I_2(\beta)/I_1(\beta)$ to 0.1σ. Right: at $\beta = 2.0$ in 4D the plaquette splits with the bare anisotropy — temporal plaquettes grow, spatial ones shrink — and a Creutz-ratio estimate gives a renormalized $\xi_R \approx 3.32$ at bare $\xi = 3$.

$\Delta \approx 3$ — the textbook heavy-0^{++} failure of Section 4, verbatim. Adding the full multi-level GEVP (Run 1) produced a flatter, lower ground state but still *no plateau*: with the signal dead by $\Delta \approx 3$ there is nothing late-Δ for any basis to exploit. The diagnosis, important because it redirects the effort: the *operator basis was not the bottleneck; the lattice was.* No amount of overlap engineering survives $m_G\, a \sim 1$ on an isotropic lattice.

7.2 Run 2: validating the anisotropic implementation

Before trusting physics from the anisotropic action (18), three checks (Figure 1):

- **$\xi = 1$ regression:** the anisotropic code path at $\xi = 1$ reproduces the exactly known 2D $SU(2)$ mean plaquette $I_2(\beta)/I_1(\beta)$: measured 0.4329 ± 0.0017 vs. exact 0.4331 (0.1σ) — the refactor changed nothing where it must not.
- **Plaquette splitting:** in 4D, $\langle P_{st} \rangle$ rises and $\langle P_{ss} \rangle$ falls with ξ (since $\beta_t > \beta_s$), coinciding at $\xi = 1$ — the anisotropy acts in the intended direction.
- **Renormalized anisotropy:** a Creutz-ratio-based estimate gives $\xi_R \approx 3.32$ at bare $\xi = 3$ — the expected tree-level vs. renormalized mismatch, made visible rather than tuned away.

7.3 Run 3: the anchor, $m\, a_t \approx 0.33$

The production ensemble is $12^3 \times 24$ at $\beta = 2.4$, bare $\xi = 3.0$, $N = 2000$ configurations (heat-bath + overrelaxation, acceptance 1.00). The anisotropy transforms the measurement (Figure 2): $C(\Delta)$ now decays cleanly over ∼10–12 timeslices before the periodic turnaround at $\Delta = N_t/2 = 12$, instead of dying at $\Delta \approx 3$. The GEVP ground state (levels $[0, 2, 4, 6]$, $t_0 = 1$) plateaus with tight errors:

$$m_{\text{eff}}(1) = 0.365 \pm 0.008, \quad m_{\text{eff}}(2) = 0.333 \pm 0.011 \qquad (\text{units of } a_t), \tag{19}$$

a small descent then flattening — the expected approach from above of the variational bound. Because two points that differ by $\sim 2\sigma$ are a descent, not yet a plateau, the anchor was later

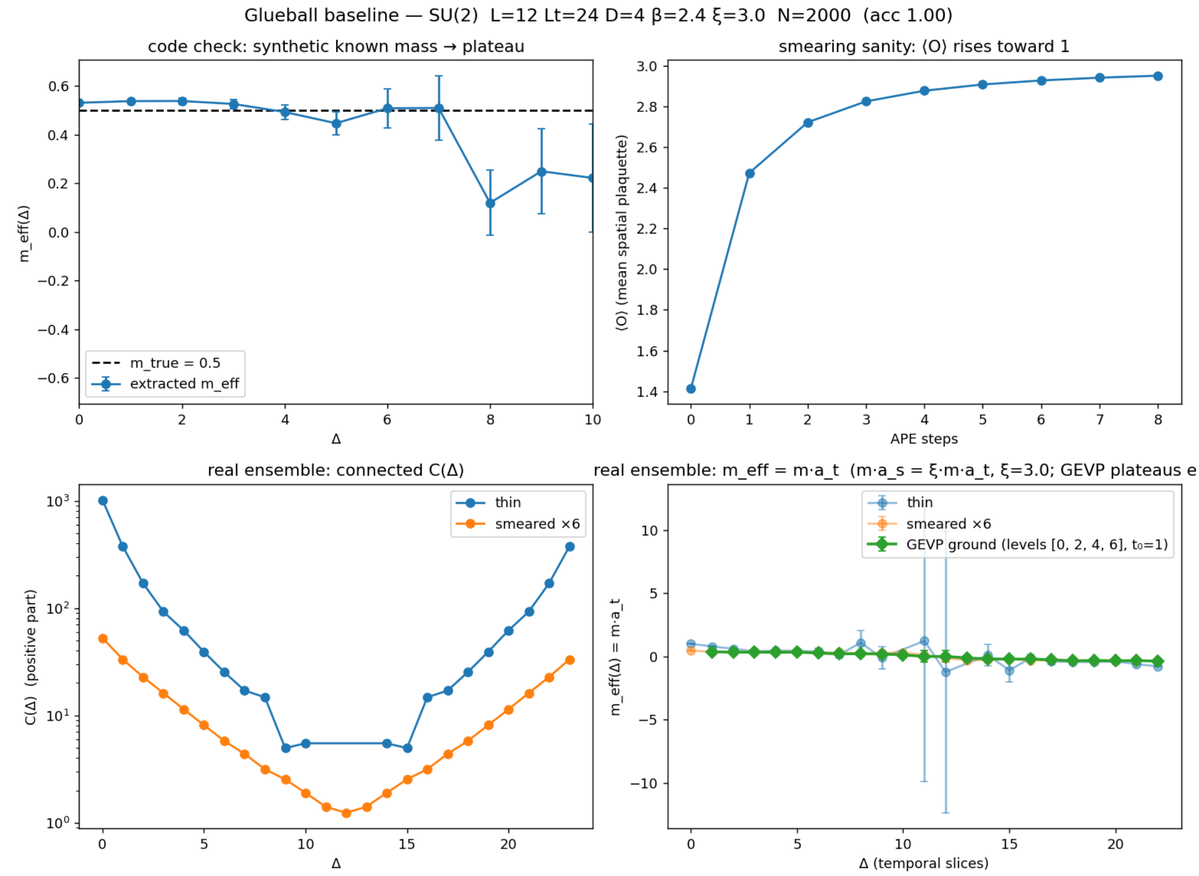

Figure 2: The classical baseline on the anisotropic ensemble ($12^3 \times 24$, $\beta = 2.4$, $\xi = 3$, $N = 2000$). Top left: the analysis code recovers a known mass from a synthetic single-exponential correlator. Top right: the smeared operator's expectation value grows monotonically with APE steps (smearing sanity check). Bottom left: the connected correlator $C(\Delta)$ for thin and 6×-smeared operators decays over ∼10–12 slices, symmetric about $\Delta = N_t/2$ (periodicity). Bottom right: effective masses — the multi-level GEVP ground state (green) plateaus where the single operators do not.

strengthened with $\Delta = 3$–4 points on the held-out test split of the same ensemble (Section 9.1): $m_{\text{eff}}(3) = 0.333 \pm 0.036$ confirms

$$\boxed{m\,a_t \approx 0.33} \tag{20}$$

as the classical ground-truth anchor for everything that follows.

7.4 What the number does and does not mean

$m\,a_t \approx 0.33$ is the trustworthy, method-validating result. The conversion $m\,a_s = \xi\,m\,a_t \approx 1.0$ is *not* continuum physics: at $\beta_s = \beta/\xi = 0.8$ the spatial lattice is deep in strong coupling (coarse a_s), so $m\,a_s$ carries large discretization artifacts, and the conversion moreover uses the bare $\xi = 3$ where Run 2 measured $\xi_R \approx 3.32$. A publishable m_G would require tuned anisotropy (raising β with ξ to keep β_s in the scaling window) and a continuum extrapolation — deliberately left as future work. The claim established here is the one the rest of this note needs: *the pipeline resolves a plateau*, i.e. a classical ground truth exists on this ensemble.

8 A neural network as a variational operator

Everything so far is a search for the operator with maximal ground-state overlap, conducted by hand: choose loop shapes, choose smearing levels, let a GEVP take one global linear combination. The observation driving the second half of this note is that this search is an *optimization problem with a differentiable objective*, and the objective is exactly a physics quantity.

8.1 The Rayleigh-quotient loss

Let $O_\theta(\vec{x}, t)$ be *any* gauge-invariant scalar field built from the spatial links of timeslice t (with parameters θ), and $\bar{O}_\theta(t)$ its zero-momentum projection. Write $|\psi\rangle = (\hat{\bar{O}}_\theta - \langle \bar{O}_\theta \rangle)\,|0\rangle$ for the vacuum-subtracted state it creates. Then by construction $C(\Delta) = \langle\psi|\hat{T}^\Delta|\psi\rangle$ and $\langle\psi|0\rangle = 0$, so the correlator ratio

$$R(\theta) \;=\; \frac{C(1)}{C(0)} \;=\; \frac{\langle\psi|\hat{T}|\psi\rangle}{\langle\psi|\psi\rangle} \tag{21}$$

is a *Rayleigh quotient of the transfer matrix* restricted to the orthogonal complement of the vacuum. Since $\hat{T}$ is positive and self-adjoint (reflection positivity again), the supremum of (21) over all operators is the largest non-vacuum eigenvalue,

$$R(\theta) \;\leq\; e^{-m_G\, a_t}, \tag{22}$$

with equality exactly when $|\psi\rangle$ is the glueball ground state. Therefore:

> **Train a network on the loss $\mathcal{L}(\theta) = -\,C(1)/C(0)$.** The loss is a rigorous variational bound; at convergence $m_G\, a_t = -\log R$; the converged loss value *is* the glueball mass, and the trained network *is* the optimal interpolating operator.

No labels, no supervision — the training signal is the same Monte Carlo correlator the classical analysis measures. Two structural gifts: the quotient is invariant under $O \to \lambda O$, so the output needs no normalization (this gift has a sting; Section 8.4); and, as in Section 3, the 0^{++} channel needs no quantum-number projection for the ground state. In practice the loss used below averages two ratios, $\mathcal{L} = -\frac{1}{2}\,[\,C(1)/C(0) + C(2)/C(0)\,]$, whose floor at the anchor mass is $-\frac{1}{2}(e^{-ma_t} + e^{-2ma_t}) \approx -0.62$ at $m\, a_t = 0.33$ — a number to remember.

8.2 The GELT architecture in brief

The operator family O_θ is GELT, a gauge-equivariant lattice transformer in the spirit of the L-CNN framework [14]. Only its contract matters here:

- **Input:** untraced plaquettes $W(x)$, which transform *adjointly*, $W(x) \to \Omega(x)W(x)\Omega^\dagger(x)$, plus parallel transporters $T_{\Delta x}(x)$ to every site in an L^1-ball of radius R, averaged over all shortest lattice paths.

- **Attention:** scores are gauge *invariant* $(\mathrm{Re}\,\mathrm{Tr}[Q^\dagger(x)\,\tilde{K}(x{+}\Delta x)]$, with K transported to x), so the attention map itself is a gauge-invariant, physically interpretable object.

- **Values:** the value path is matrix-*bilinear*, $\sum \alpha\, Q_v^\dagger \tilde{V}$ — each layer multiplies transported loop matrices, so depth composes ever-larger Wilson loops, the same universality mechanism as L-CNN's bilinear layers.

- **Output:** a trace and per-site MLP produce a gauge-invariant scalar field $O_\theta(x)$ per site — precisely an operator density.

Equivariance throughout guarantees the output is exactly gauge invariant, so O_θ is a legitimate physical operator for any θ: the variational bound of Section 8.1 holds at every step of training, not just at convergence. Conceptually, GELT is a *learned, content-dependent smearing*: transport-averaging plus stacked attention is a gauge-covariant multi-scale aggregation, and the Rayleigh loss tunes it for ground-state overlap directly — the thing hand-tuned APE schedules approximate.

8.3 The single-timeslice constraint (where the theory bites)

The derivation of (21) required $\bar{O}(t)$ to be a functional of the fields on *timeslice t only* — the same rule that forbids temporal smearing (Section 5.1), now applied to the network itself. A 4D network violates it doubly: temporal plaquettes among its input channels, and a receptive field (transport radius × depth) that leaks across timeslices. The failure is not a small correction; it makes the loss *gameable*. If $O(t)$ may depend on neighbouring slices, the optimizer's cheapest route to $C(1)/C(0) \to 1$ is an output that varies slowly in t — in the limit, a per-configuration constant gives $C(\Delta) = C(0)$ for all Δ and a fake mass of zero. The network would "beat" every classical operator for a purely unphysical reason.

The fix defines the setup: **run GELT as a three-dimensional network on the spatial links of one timeslice**, batched over (configuration × timeslice), with per-timeslice outputs reshaped to $\bar{O}(t)$ for the correlator. Within its slice the operator keeps its full spatial loop-building power — exactly the domain the classical operator uses. The absence of temporal loop content is not a limitation; it is the definition of a valid interpolating operator. (The 3D restriction also collapses the transport memory cost by $\sim 70\times$, making on-the-fly transport per batch cheap — the constraint that protects the physics also pays the compute bill.)

8.4 Making the loss trainable

Four practical pathologies, each discovered the hard way:

Zero gradient at initialization. The network's readout was zero-initialized (a common stabilization). But $C(0)$ and $C(1)$ are both quadratic in the output, so at $O \equiv 0$ the gradient of $-C(1)/(C(0)+\epsilon)$ is *exactly* zero — training never leaves the saddle. The readout must be initialized nonzero.

Denominators. An early loss chained ratios $C(\Delta)/C(\Delta-1)$, putting the small, signed, noisy $C(1)$ in a denominator; clamping kept the values finite but the *gradients* were garbage in direction, and the operator scale was pumped to $C(0) \sim 5 \times 10^8$. Every ratio now has the large, positive $C(0)$ as its denominator, and the correlator arithmetic runs in float64.

The scale-flat direction. Scale invariance of the quotient means θ-space has an exactly flat direction $O \to \lambda O$. With smooth, site-coherent inputs (Section 9.2) the bilinear value path squares its gain per layer, and the optimizer drifted along the flat direction to $C(0) \sim 10^{73}$ — float32 overflow, `NaN` validation loss, and (insidiously) no checkpoint ever saved, because `NaN` never improves the best-so-far. The cure is a variationally neutral *scale pin*: adding $(\log C(0))^2$ to the loss penalizes only the flat direction (it is minimized at $C(0) = 1$ and does not touch the ratio), removing the runaway without biasing the physics.

Estimator bias and overfitting. The loss is a ratio of *batch-estimated* means (including the vacuum subtraction, itself a noisy batch estimate), hence biased on small batches — so batches are made as large as memory allows and the final numbers never come from the training estimator. More fundamentally, the variational bound holds in *expectation*, not on a finite sample: a network can overfit the noise of its training correlator and spuriously "violate" the bound there. The protocol is therefore a hard split (70% train / 10% validation / 20% test, contiguous in Monte Carlo time), checkpoint selection on validation, and every reported mass from a blocked jackknife on the untouched test configurations only.

9 Training runs: from a failed operator to one that wins

All runs train on the cached Run-3 ensemble ($12^3 \times 24$, $\beta = 2.4$, $\xi = 3$, $N = 2000$; 1400 train / 200 validation / 400 test configurations), with a small per-timeslice 3D GELT ($R = 2$, 4 layers, 2 heads, $d_{\text{qkv}} = 6$, $d_{\text{model}} = 16$; $\sim$5k parameters), AdamW with cosine schedule, batches of 6 configurations (= 144 timeslices), and gradient checkpointing on a V100. The classical GEVP of Section 7, re-evaluated on the same test split with the same blocked jackknife, is the baseline throughout.

9.1 Run 4 — thin input: a genuine operator that loses on overlap

Fed thin spatial plaquettes, GELT trains stably to a best validation loss of -0.433 — well short of the ≈ -0.62 variational floor — with a clear overfitting signature (validation bottoms out near epoch 12 while the training loss keeps falling; Figure 3, left). On test:

operator	$m_{\text{eff}}(1)$	$m_{\text{eff}}(2)$	$m_{\text{eff}}(3)$	$m_{\text{eff}}(4)$
GELT (thin input)	0.527 ± 0.028	0.441 ± 0.042	0.310 ± 0.045	~ 0.5, large bars
classical GEVP	0.398 ± 0.016	0.357 ± 0.025	0.333 ± 0.036	0.284 ± 0.050

Table 1: Run 4 (thin-input GELT) vs. the classical GEVP, blocked jackknife on the 400 test configurations. Units of a_t.

The learned operator is *genuine* — its m_{eff} descends $0.705 \to 0.527 \to 0.441 \to 0.310$, the $\Delta = 3$ point consistent with the anchor, and it clearly beats the thin classical operator (≈ 0.8 at $\Delta = 1$) with no hand-built smearing. But it carries excited-state contamination the smeared GEVP basis has projected out, and no convincing plateau of its own.

The diagnosis is a *depth budget.* Four bilinear layers compose loops of degree ≤ 16 in the input plaquettes; six iterated APE steps contain branched staple content of degree $\sim 3^6$ in the links within the same spatial radius. The network cannot re-derive an iterated smearing schedule at affordable depth. This diagnosis was made quantitative by an eval-only diagnostic (Run 4b): the correlation of the learned $\bar{O}$ with the APE ladder on test fluctuations peaks at APE$\times$2 ($|r| = 0.68,\ \mathbf{0.87},\ 0.71,\ 0.59$ for levels $0, 2, 4, 6$) — the depth-4 network learned about two APE steps' worth of staple content. Run 4b also appended the learned operator to the classical basis as a fifth GEVP operator: the enlarged basis is *identical* to the classical one within small fractions of the error bars (Figure 3, the overlapping curves). The $\sim 25\%$ of the learned operator's variance that is linearly independent of the ladder carries no additional ground-state overlap — the thin-input operator is variationally redundant. This null is the clean baseline for what follows: any future separation of the enlarged GEVP from the classical curve must come from what the network computes *on top of* smearing, not from smearing itself.

9.2 Run 5 — smeared input channels: the physics fix

If depth cannot rebuild iterated smearing, give the network the smearing: feed plaquettes built at the same cumulative APE levels the GEVP uses $(0, 2, 4, 6)$, stacked as input channels (3 spatial planes $\times$ 4 levels = 12 channels), with parallel transport still built from thin links. Smearing is spatial-only, so the single-timeslice transfer-matrix condition is untouched. The

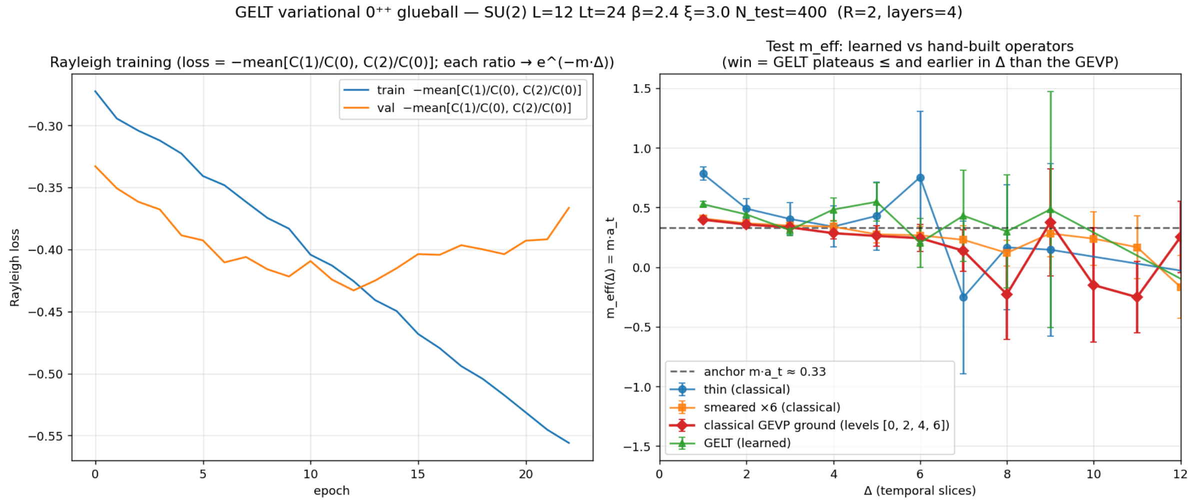

Figure 3: Run 4: GELT trained from thin spatial plaquettes. Left: Rayleigh training curves; validation (orange) bottoms out near epoch 12 and turns up — overfitting of the empirical correlator — while the best value -0.433 stays far from the ≈ -0.62 variational floor. Right: test effective masses. The learned operator (green) is genuine (descending toward the anchor, far better than the thin operator) but sits above the classical GEVP (red) at small Δ: an overlap gap.

comparison also becomes maximally sharp: both methods now see *exactly the same* smearing levels — the GEVP may take one global linear combination of them, while GELT may mix them nonlinearly and site-by-site. Any gap is attributable to the network.

The first attempt hit the scale-flat runaway of Section 8.4 (smooth coherent channels $\rightarrow$ constructive attention sums $\rightarrow$ squared per-layer gain $\rightarrow C(0) \sim 10^{73}$ by epoch 2); with the $(\log C(0))^2$ scale pin the warm-started retrain is stable throughout — $C(0)$ held at $\mathcal{O}(1)$, no overfit turn-up, validation flat on the floor — reaching best validation loss

$$\mathcal{L}_{\text{val}} = -0.6185 \quad \Longleftrightarrow \quad m\, a_t \approx 0.329, \tag{23}$$

i.e. the Rayleigh quotient *saturates the transfer-matrix bound at the anchor mass*: the converged loss is the glueball mass, realized. On test (Figure 4):

operator	$m_{\text{eff}}(1)$	$m_{\text{eff}}(2)$	$m_{\text{eff}}(3)$	$m_{\text{eff}}(4)$
GELT (smeared input)	$\mathbf{0.370 \pm 0.016}$	$\mathbf{0.333 \pm 0.020}$	0.345 ± 0.029	—
classical GEVP	0.398 ± 0.016	0.357 ± 0.025	0.333 ± 0.036	0.284 ± 0.050
GEVP + GELT (5 ops)	0.366 ± 0.017	0.333 ± 0.020	0.347 ± 0.030	0.319 ± 0.048

Table 2: Run 5 final results, blocked jackknife on the 400 test configurations. Units of a_t. The learned single operator plateaus from $\Delta = 1$; the enlarged five-operator GEVP collapses onto it.

Since GELT and the GEVP are measured on the same test configurations, their error bars are correlated and comparing them independently would be dishonest. The significance test is the blocked jackknife of the *difference* $m_{\text{eff}}^{\text{GELT}} - m_{\text{eff}}^{\text{GEVP}}$, in which the shared fluctuations cancel:

$$\Delta{=}1: \ -0.028 \pm 0.007\ (3.9\sigma), \qquad \Delta{=}2: \ -0.024 \pm 0.014\ (1.8\sigma), \qquad \Delta{=}3: \ +0.012 \pm 0.021. \tag{24}$$

The learned operator sits *significantly below* the classical GEVP exactly where excited-state contamination lives (small Δ) and is statistically identical on the plateau: a better operator

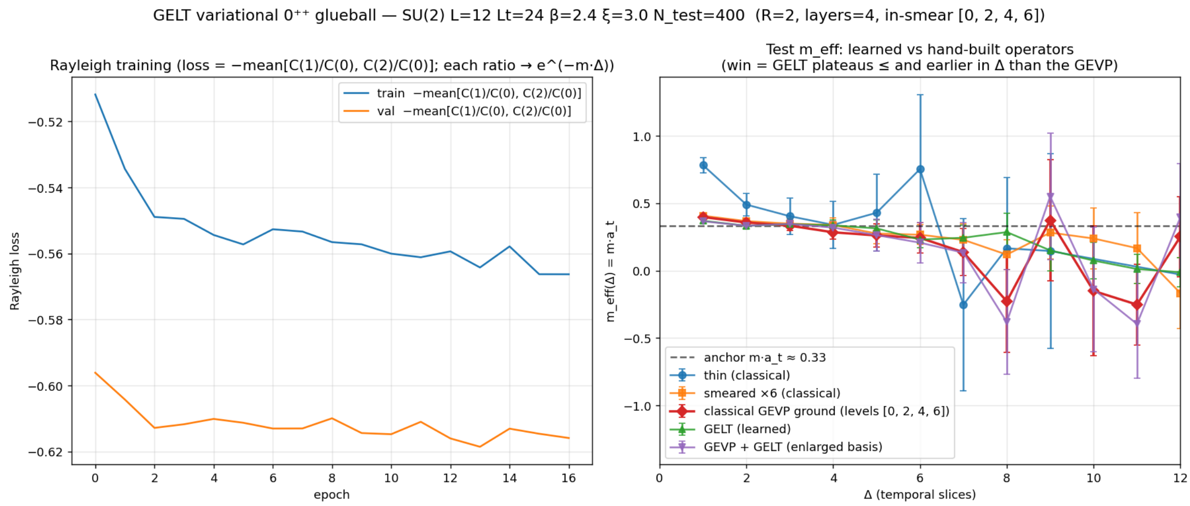

Figure 4: Run 5 (final): GELT with multi-level smeared input channels ($[0, 2, 4, 6]$). Left: with the scale pin the training is stable and the validation Rayleigh loss sits on the variational floor, $-0.6185 \leftrightarrow m\,a_t \approx 0.329$ (validation below training is an estimator artifact: the 6-configuration batch estimator is noise-biased toward zero, the 200-configuration validation estimate is cleaner). Right: test effective masses. The learned operator (green) is flat on the anchor (dashed) from $\Delta = 1$, below the still-descending classical GEVP (red) at small Δ; the enlarged GEVP + GELT basis (purple) collapses onto pure GELT.

measuring the same physics — which is precisely what the variational bound demands of a legitimate win. Three corroborating observations:

- **The win criterion is met.** GELT is flat at ≈ 0.33–0.37 from $\Delta = 1$ (in Figure 4 it hugs the anchor line through $\Delta \approx 6$) while the GEVP is still descending at $\Delta = 1$–2. Even GELT's most contaminated point, the $\Delta{=}0{\to}1$ Rayleigh mass 0.392, sits below the GEVP's $m_{\text{eff}}(1)$.
- **One learned operator subsumes the hand-built basis.** Offered the classical four-level ladder *plus* GELT, the variational solver picks essentially pure GELT — the enlarged GEVP equals GELT's own m_{eff} to within ~ 0.004 at $\Delta = 1$. Against the Run-4b null, this separation is cleanly attributable to the network's nonlinear, spatially resolved mixing of the same smearing levels.
- **The network adds content, not just smearing.** The ladder correlations of the trained operator are $|r| = 0.45, 0.76, 0.87, 0.91$ for levels $0, 2, 4, 6$ — it lives at the deep end of the ladder — and training grew the ladder-*independent* variance from $\sim 8\%$ to $\sim 18\%$ *while the loss improved*: the network's own content, not residual noise, drives the win.

9.3 Quoting the result like a lattice paper: cosh fits and the ground-state overlap A_0

Table 2 compares effective masses point by point. That is how lattice results are *plotted*, but not how they are *quoted*: the convention of the field is to quote the mass from a fit of $C(\Delta)$ to the single-state form over a plateau window, drawn as a horizontal band through the m_{eff} points, and to quote *operator quality* — which is what the GELT-vs-GEVP comparison is actually about — as the *ground-state overlap*, the number Morningstar and Peardon use to rank their smeared and fuzzed operators [12]. This subsection re-expresses the Run-5 result in that standard form (Figure 5).

Each operator's connected correlator is fitted to the periodic (finite-N_t) single-state form of Section 3,

$$C(\Delta) \;\approx\; A\,\big[\,e^{-m\,a_t\,\Delta} + e^{-m\,a_t\,(N_t-\Delta)}\,\big], \tag{25}$$

over one *shared* window $\Delta \in [2,7]$ — comparability beats per-operator optimality, and starting at $\Delta = 2$ keeps the GEVP's residual $\Delta = 1$ contamination out of *its* mass fit: the contamination is what the overlap should report, not what the fit should absorb. Extrapolating the fitted ground-state piece back to $\Delta = 0$ and comparing with the measured $C(0) = \sum_{n>0} |\langle 0|\hat{\bar{O}}|n\rangle|^2$ gives the **ground-state overlap fraction**

$$A_0 \;=\; \frac{A\,(1+e^{-m\,a_t\,N_t})}{C(0)} \;\in\; (0,1], \tag{26}$$

the fraction of the operator's spectral weight carried by the glueball; $A_0 = 1$ is a pure ground-state interpolator. This closes a conceptual loop: the Rayleigh quotient (21) is (up to the wrap term) a weighted average $\sum_n w_n\, e^{-m_n a_t}$ with weights summing to one and $w_{\text{ground}} = A_0$, so saturating the variational bound *is* the statement $A_0 \to 1$ — the quantity quoted here is the quantity the training maximizes. Objective and metric coincide.

The comparison also needs the right comparator. A single learned operator should not be measured against a whole basis but against the best *single* operator the basis contains; the GEVP supplies exactly that (Section 5.2): the ground-state eigenvector v_0 of (16), solved once at reference times $(t_0, t_d) = (1,2)$, defines the fixed-vector *projected operator* $\bar{O}_{\text{proj}}(t) = \sum_i (v_0)_i\, \bar{O}_i(t)$ — the variational optimum in the span of the four smearing levels [11] — whose scalar correlator is fitted like any other. Errors follow the same discipline as before: the σ_Δ entering the (diagonal) χ^2 are fixed from the full-sample blocked jackknife, the *entire* fit — including v_0 — is redone inside each of the 40 delete-block samples, and the GELT−GEVP differences are jackknifed directly so the shared-ensemble fluctuations cancel. On the 400 test configurations:

operator	$m\,a_t$ (fit)	A_0	χ^2/dof
GELT (learned)	0.332 ± 0.027	$\mathbf{0.903 \pm 0.047}$	0.05
GEVP-projected	0.340 ± 0.030	0.837 ± 0.056	0.04
APE×6	0.340 ± 0.032	0.805 ± 0.052	0.11

Table 3: Cosh fits (25) on the shared window $\Delta \in [2,7]$, blocked jackknife on the 400 test configurations (dof = 4). The χ^2/dof values are artificially low because neighbouring $C(\Delta)$ points are strongly correlated while the χ^2 is diagonal — a fit-quality heuristic, not a goodness-of-fit test; the quoted errors come from the jackknife, which propagates the correlations correctly.

with correlated differences

$$\Delta(m\,a_t) \;=\; -0.008 \pm 0.014 \;\;(0.6\sigma), \qquad \Delta A_0 \;=\; +0.066 \pm 0.031 \;\;(2.1\sigma). \tag{27}$$

The headline, in the field's own language: *both operators measure the same mass, but the learned operator carries* 90% *of its spectral weight on the ground state against* 84% *for the optimal combination of the entire classical basis.* Two readings of Table 3 sharpen it:

- **The GEVP barely improves on its best member; GELT does.** The optimal classical combination gains only ~ 0.03 of overlap over APE×6 alone (0.837 vs 0.805), while GELT gains ~ 0.10 over the full basis: one learned operator adds more ground-state overlap than the whole Morningstar–Peardon construction adds over its best single level — the fit-language restatement of the enlarged-GEVP collapse of Section 9.2.

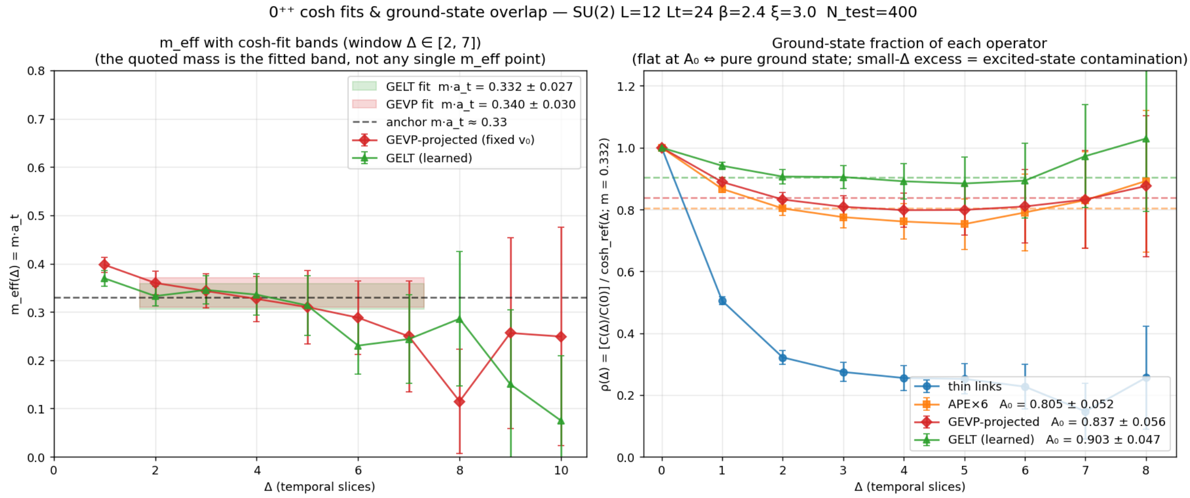

Figure 5: The Run-5 result in standard spectroscopic form. Left: test $m_{\text{eff}}(\Delta)$ for GELT (green) and the projected GEVP operator (red) with their fitted-mass bands over the shared window $\Delta \in [2, 7]$ — the quoted mass is the band, not any single point; both bands sit on the classical anchor (dashed). Right: the ground-state fraction $\rho(\Delta) = [C(\Delta)/C(0)]\,/\,\cosh_{\text{ref}}(\Delta)$, with $\cosh_{\text{ref}}$ the fitted GELT ground state normalised to $\rho(0) = 1$. A pure ground-state interpolator is flat at its A_0 (dashed lines); excited-state contamination is the excess at small Δ. Thin links collapse to $\rho \approx 0.25$ immediately; APE×6 and the projected GEVP descend through $\Delta \approx 3$ toward $A_0 \approx 0.80$–0.84; GELT is flat at $A_0 \approx 0.90$ from $\Delta = 1$. (The classical curves' dip slightly below their A_0 lines near $\Delta \approx 4$–5 is noise plus the cosh wrap, well within the error bars — an exact spectral sum would approach A_0 from above.)

- **2.1σ and 3.9σ are complementary, not in tension.** The 3.9σ of Section 9.2 lives entirely at $\Delta = 1$, where the contamination is largest; the shared fit window starts at $\Delta = 2$, so ΔA_0 never sees GELT's cleanest advantage and reconstructs it only by extrapolation. A_0 is the *physical* operator-quality statement; the correlated m_{eff} difference at $\Delta = 1$ remains the sharpest single-point evidence of the same fact.

9.4 Replication on an independent ensemble

Everything so far is one ensemble. To close that caveat, the entire chain — heat-bath/overrelaxation sampling, from-scratch training, checkpoint selection, test-split analysis — was repeated with a different sampler seed: a second, fully independent $12^3 \times 24$, $\beta = 2.4$, $\xi = 3$, $N = 2000$ ensemble, the same network and hyperparameters, no reuse of the Run-5 weights. Nothing about the procedure was tuned on the new data.

The replication reproduces the operator-quality result — and sharpens a point about what does *not* replicate. On the new ensemble every operator, classical and learned alike, reads $\sim 1\sigma$ heavier: the classical GEVP sits flat at $m_{\text{eff}} \approx 0.38$–$0.40$ rather than descending through 0.36 toward 0.33, and the fitted masses come out 0.37–0.40. Correspondingly, training converged to a best validation Rayleigh loss of -0.564 rather than -0.619 — which decodes to $m \approx 0.395$, i.e. the variational floor *of this ensemble's own correlator*, saturated to ~ 0.004 exactly as before. The Rayleigh floor is an empirical property of the ensemble, not a universal target: what the loss saturates is the transfer-matrix bound as realized on the sampled data, so the mass fluctuates between independent ensembles (here by 0.04 ± 0.04, an ordinary $\sim 1\sigma$ statistical difference on 2000 configurations — the "anchor" carries its own ensemble error), while the operator-quality statements are what should — and do — replicate. On the new test split (400 configurations, same shared window $\Delta \in [2, 7]$):

operator	$m\,a_t$ (fit)	A_0	χ^2/dof
GELT (learned)	0.374 ± 0.031	$\mathbf{1.013 \pm 0.062}$	0.01
GEVP-projected	0.400 ± 0.038	0.925 ± 0.071	0.08
APE×6	0.390 ± 0.037	0.901 ± 0.068	0.05

Table 4: The replication ensemble's analogue of Table 3: cosh fits on the shared window $\Delta \in [2, 7]$, blocked jackknife on the 400 test configurations. A_0 exceeds 1 only within its error — consistent with a pure ground-state interpolator.

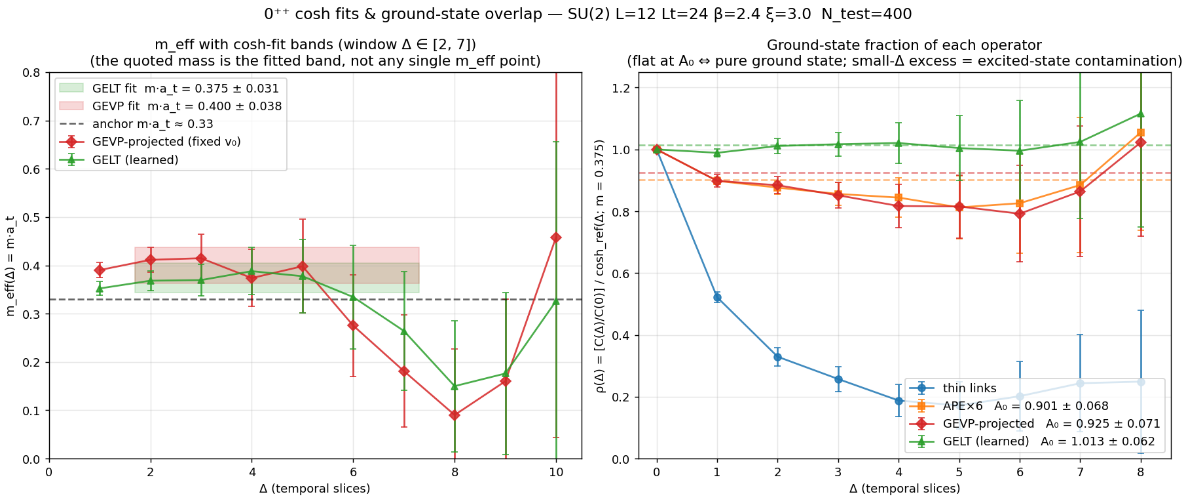

Figure 6: Figure 5 repeated on the independent replication ensemble: $m_{\text{eff}}(\Delta)$ with fitted-mass bands (left) and the ground-state fraction $\rho(\Delta)$ (right). The overall mass scale sits $\sim 1\sigma$ higher than on the original ensemble — both operators agree on that shift, as they must — while the operator-quality ordering (GELT flat at its A_0 from small Δ, the classical operators descending into it) is unchanged.

with correlated differences

$$\Delta(m\,a_t) \;=\; -0.025 \pm 0.013 \;\; (1.9\sigma), \qquad \Delta A_0 \;=\; +0.089 \pm 0.030 \;\; (2.9\sigma), \tag{28}$$

and a correlated m_{eff} difference of -0.038 ± 0.008 (4.5σ) at $\Delta = 1$ and -0.036 ± 0.012 (2.9σ) at $\Delta = 2$ — the Run-5 pattern, if anything stronger. The enlarged GEVP again collapses onto the learned operator (the classical ladder adds ~ 0.005 at $\Delta = 1$, within noise), and the smearing-ladder correlations are nearly identical ($r = 0.36/0.75/0.89/0.93$ against APE×0/2/4/6).

Since the two ensembles are statistically independent, the two overlap differences combine: inverse-variance weighting of $+0.066 \pm 0.031$ (Run 5) and $+0.089 \pm 0.030$ (replication) gives

$$\Delta A_0^{\text{combined}} \;=\; +0.078 \pm 0.022 \qquad (3.6\sigma). \tag{29}$$

The learned operator's excess ground-state overlap over the optimal classical combination is not a feature of one ensemble.

10 What the learned operator attends to

The validation above establishes *that* the learned operator is better; it says nothing about *how*. Because every GELT attention score is a gauge invariant — Re $\mathrm{Tr}[Q^\dagger \tilde{K}]$ between parallel transported adjoint fields — the attention weights $\alpha_n(x)$ are not an implementation detail but a measurable field: a per-site probability distribution over the 24 neighbours in the L_1 ball of radius $R = 2$, plus the site itself. This section reads that distribution out of the Run-5 checkpoint and asks what structure the optimal operator has learned. It is a first look, on deliberately modest statistics, not a finished study.

Setup. A fresh $12^3 \times 24$, $\beta = 2.4$, $\xi = 3$ ensemble of $N = 32$ configurations was sampled with a new seed, so it is held out by construction; the Run-5 checkpoint was loaded unchanged and run in inference over all $32 \times 24 = 768$ timeslices. Three statistics are accumulated per (layer, head): the *attention range* $\ell_{\text{att}}(x) = \sum_n \alpha_n(x)\,|\Delta x_n|_1$, the offset entropy $-\sum_n \alpha_n \log \alpha_n$ (maximal value $\log 25 = 3.22$), and a per-site axial anisotropy — the relative spread, across the three spatial axes, of the attention mass at $|\Delta x|_1 = 1$ (maximal value $\sqrt{2} = 1.41$, attained when all of it sits on one axis). Two further arms follow the rule that a correlation is not an explanation: the per-timeslice Spearman correlation of $\ell_{\text{att}}(x)$ against the smeared spatial-plaquette density $o(x)$ (the physical field this operator measures — large o means *small* Wilson action density, since $S \propto \sum_x (1 - o(x)/3)$), and a head-ablation intervention in which head h's column of layer l's output mixing matrix is zeroed and the degradation in the Rayleigh loss measured. Errors on both validation arms respect the correct independent unit: the ablation Δloss is a delete-one jackknife over configurations of the *correlated* difference (ablated and intact operators evaluated on the same configurations, so the shared ensemble noise cancels), and the Spearman error is the spread of per-configuration means, not of per-slice values — slices within a configuration are correlated over $\sim 1/m$, which is the very quantity this report measures. The whole study runs in about an hour on a laptop CPU.

The checkpoint transfers. On this previously unseen chain the operator returns $C(1)/C(0) = 0.648$, i.e. a Rayleigh loss of -0.538 and $m\,a_t \approx 0.43$. This is a third independent realization of the ensemble-specific variational floor discussed in Section 9.4 (0.33 on Run 5, 0.395 on the replication ensemble), and it confirms that the trained weights are not tuned to their training chain. The number should be read with its statistical width in mind: the same quantity on the first 8 configurations of this same chain reads $m\,a_t \approx 0.375$, so a Rayleigh ratio on tens of configurations carries an uncertainty of order 0.05 — which is exactly why the intervention arm below needed error bars before it could be interpreted at all.

One head does the work. The ablation is strikingly concentrated: head 1 of layer 3 costs $+0.0787 \pm 0.0145$ in Rayleigh loss when removed — 5.4σ, an order of magnitude more than any other head, and roughly a seventh of the entire signal — while no other head reaches 2σ in the positive direction. That same head is the only *local* one in the network: $\ell_{\text{att}} = 0.30$ with 71% of its attention mass on the site itself, against six heads that sit at or near the maximal range $\ell_{\text{att}} = 2$. The optimal operator is therefore not a democratic ensemble of scales but one content-dependent local head embedded in a stack of long-range ones.

Two heads carry a mildly *negative* Δloss (layer 1 head 0 and layer 2 head 0, both ≈ -0.0024 at 2.2–2.3σ): removing them marginally improves the Rayleigh ratio on this ensemble. Eight cells were tested, so ~ 0.2 excursions beyond 2.2σ are expected by chance and two is only a mild excess; but the direction is also what mild overfitting would produce, since these weights were optimized on the Rayleigh ratio of a *different* ensemble and are being scored here on a fresh one. Either way the effect is $30\times$ smaller than the layer-3 head and does not bear on the operator's measured quality.

No emergent coarsening. The interpretability program's Study-1 hypothesis — that a learned operator should widen its receptive field with depth, an RG-like coarsening no fixed architecture imposes — is *not* supported here. ℓ_{att} is essentially flat across layers at the ceiling $\ell = 2$, and the one departure (layer 3, head 1) goes the wrong way. Worse, two heads are

layer	head	ℓ_{att}	entropy	anisotropy	axis split	Spearman(ℓ, o)
1	0	1.69 ± 0.15	2.24	0.56	0.40/0.31/0.29	−0.139(1)
1	1	1.87 ± 0.06	1.04	0.99	0.00/1.00/0.00	−0.009(1)
2	0	1.93 ± 0.05	0.35	1.37	0.00/1.00/0.00	−0.086(1)
2	1	2.00 ± 0.00	0.004	1.41	1.00/0.00/0.00	+0.397(2)
3	0	2.00 ± 0.01	0.07	1.41	0.00/1.00/0.00	−0.088(1)
3	1	$\mathbf{0.30 \pm 0.18}$	0.89	0.79	0.81/0.00/0.19	−0.328(2)
4	0	1.59 ± 0.21	1.25	1.36	0.99/0.00/0.00	−0.230(2)
4	1	1.44 ± 0.12	2.22	0.75	0.00/0.64/0.36	**−0.564(2)**

layer	head	Δloss (ablate)	significance
1	0	-0.0023 ± 0.0010	-2.3σ
1	1	-0.0002 ± 0.0005	-0.4σ
2	0	-0.0025 ± 0.0011	-2.2σ
2	1	-0.0003 ± 0.0017	-0.2σ
3	0	-0.0010 ± 0.0014	-0.7σ
3	1	$\mathbf{+0.0787 \pm 0.0145}$	$\mathbf{+5.4\sigma}$
4	0	$+0.0005 \pm 0.0055$	$+0.1\sigma$
4	1	$+0.0058 \pm 0.0036$	$+1.6\sigma$

Table 5: Attention statistics (top) and head-ablation intervention (bottom) of the Run-5 operator on the held-out seed-7 ensemble (32 configurations, 768 timeslices, 1728 sites each). The ℓ_{att} error is the *site-to-site* spread, not a mean error: it is the configuration-dependent part of the attention, the part a fixed convolution kernel could not supply. "Axis split" is the fraction of sites at which the head selects each spatial axis for its $|\Delta x|_1 = 1$ attention mass. Spearman errors are configuration-level. Baseline Rayleigh loss −0.538; a positive Δloss means removing the head makes the operator worse, and the error is a delete-one jackknife over configurations of the correlated difference.

effectively degenerate: layer 2 head 1 has entropy 0.004 and $\ell_{\text{att}} = 2.000 \pm 0.000$, and layer 3 head 0 is close behind, meaning they place all of their mass deterministically on the outer $|\Delta x|_1 = 2$ shell regardless of the configuration. Those heads are fixed kernels, not measurements: whatever they contribute, a convolution could have supplied it. This is a negative result about the current architecture — with $R = 2$ the ball is too small for a range hierarchy to have anywhere to go — and it is worth reporting as such, since it identifies R and the number of layers, not the attention mechanism, as what to vary next.

Attention reaches further where the action is dense. The ground-truth arm is the more encouraging one. Layer 4 head 1 — the highest entropy head in the network, i.e. the most genuinely distributed — has $\ell_{\text{att}}(x)$ anticorrelating with the smeared plaquette density at Spearman -0.56. Since large $o(x)$ is *small* action density, the sign says the head extends its reach precisely on the disordered, high-action regions and contracts on smooth vacuum. That is the behaviour one would want of a glueball operator, and it is measured against a physical field rather than asserted from a picture. Layer 3 head 1, the load-bearing local head, shows the same sign more weakly (-0.33).

The axial anisotropy is a globally fixed axis. Four heads sit at anisotropy $\gtrsim 1.36$ out of a maximum of 1.41: at each site, their distance-one attention is concentrated on a single spatial axis. The 0^{++} channel is a cubic-group scalar, and nothing in GELT enforces rotational symmetry (only gauge equivariance; the shortest-path transport average is rotation-symmetric, the learned weights are not). The question is whether the selected axis varies from site to site, in which case isotropy is restored in the zero-momentum sum, or is globally fixed, in which case the operator is not a cubic-group scalar. The axis-split column settles it, and the answer is the unfavourable one: layer 1 head 1, layer 2 head 0 and layer 3 head 0 select the same axis at 100.0% of all 1.3 million site–slice samples, layer 2 head 1 selects a different single axis at 100.0%, and layer 4 head 0 at 99%. These are not content-driven choices; they are frozen directions.

Two observations keep this from being fatal. First, the axis-locked heads are exactly the low-entropy, maximal-range, ablation-irrelevant ones — the degenerate kernels of the previous paragraph — while the two heads that matter physically are the two most isotropic ones: the load-bearing layer-3 head splits 0.81/0.00/0.19 and the best-correlated layer-4 head splits 0.00/0.64/0.36. Second, a spin-2 admixture in the interpolating operator contaminates the correlator with *heavier* states, biasing an extracted mass upward, not downward; it cannot manufacture the ground-state overlap excess of Section 9.3. What it does mean is that the learned operator is a 0^{++} interpolator only approximately, and that explicitly projecting onto the cubic A_1^{++} representation — averaging the operator over the 24 lattice rotations, or symmetrizing the architecture — is a concrete and likely profitable next step.

Attention range versus correlation length: a measurement this lattice does not permit. The natural companion to the localization question is whether the learned attention range tracks a physical length — whether ℓ_{att} grows with the correlation length ξ as the coupling is varied. The relevant length is the *spatial* one, in the units the attention offsets are counted in:

$$\xi_s = \frac{1}{m\, a_s} = \frac{1}{\xi\, m\, a_t}, \tag{30}$$

since ℓ_{att} is built from spatial displacements $|\Delta x|_1$ measured in a_s. Before training one operator per coupling, it is worth asking whether ξ_s moves at all. It is a purely classical question — one ensemble and one GEVP per β, no network — and the answer decides whether the study is possible:

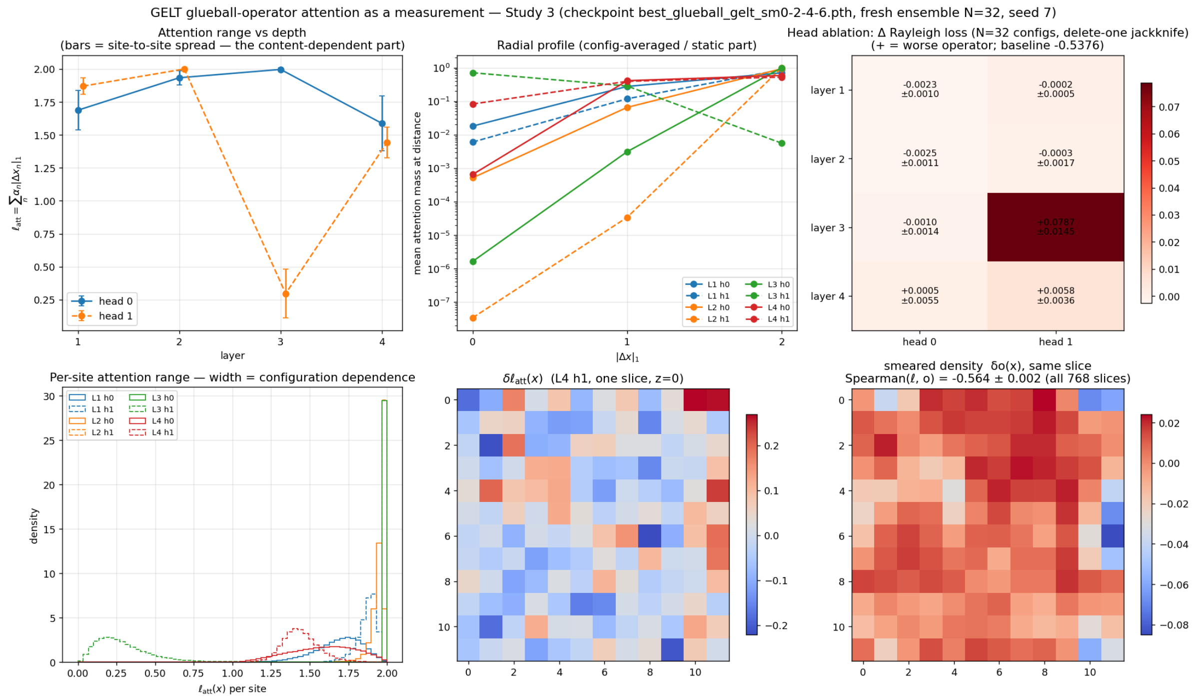

Figure 7: Attention of the Run-5 operator as a measurement. Top: ℓ_{att} versus depth with site-to-site spread as error bars (left), the config-averaged radial profile (centre), and the head-ablation matrix (right). Bottom: the distribution of the per-site range, whose width is the configuration dependence (left), and one $z = 0$ slice of $\delta\ell_{\text{att}}(x)$ next to the smeared density $\delta o(x)$ for the most strongly correlated head (centre, right).

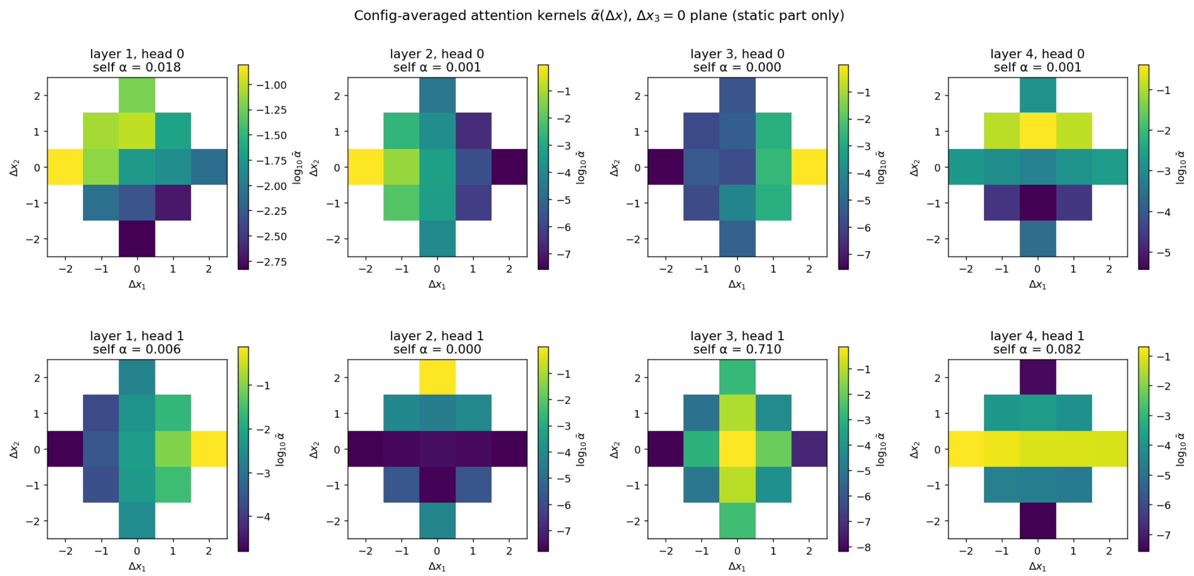

Figure 8: Config-averaged attention kernels $\bar{\alpha}(\Delta x)$ on the $\Delta x_3 = 0$ plane, per (layer, head), on a logarithmic scale. This is the *static* part of the attention only — the part expressible as a fixed convolution; the claim of Table 5 lives in the site-to-site residual around these means.

β	$\beta_s = \beta/\xi$	$m\,a_t$ (GEVP, $\Delta = 2$)	$m\,a_s$	ξ_s
2.1	0.70	0.5812 ± 0.0246	1.744	0.57 ± 0.02
2.3	0.77	0.4009 ± 0.0128	1.203	0.83 ± 0.03
2.4	0.80	0.3330 ± 0.0106	0.999	1.00 ± 0.03
2.5	0.83	0.3026 ± 0.0105	0.908	1.10 ± 0.04
2.7	0.90	0.3325 ± 0.0113	0.998	1.00 ± 0.03

Table 6: Classical β-scan on the $12^3 \times 24$, $\xi = 3$ lattice family ($N = 2000$ per coupling, multi-level GEVP, blocked jackknife). The $\beta = 2.4$ row is the anchor ensemble of Section 9 and reproduces its established value 0.333 ± 0.011, which validates the scan. ξ_s is Eq. (30), in units of the spatial lattice spacing.

The decisive number is not the span but the magnitude: ξ_s never exceeds 1.1 lattice spacings anywhere in the window. The attention offsets are *integers*, $|\Delta x|_1 \in \{0, 1, 2, \dots\}$, so the entire variation of the quantity ℓ_{att} would have to track lives inside the first offset shell. A factor of two in a sub-lattice length is not a factor of two in anything the network can represent, and training one operator per β would return a flat scatter whose flatness measured the lattice spacing rather than the attention.

It is worth being precise about what is and is not small here, because the two are easily confused. The *box* is large: $L/\xi_s \approx 12$ correlation lengths across, which is the regime in which finite-volume effects are absent, not present. What is coarse is the *spacing.* At $\xi = 3$ the spatial coupling is $\beta_s = \beta/\xi \approx 0.8$ — strong coupling — so a_s is large and the entire physical extent of the glueball falls inside a single grid cell. The lattice has plenty of cells; the object occupies about one of them. Refining a_s, not enlarging L, is what the measurement would need.

One caveat on the high-β end. $m\,a_t$ falls monotonically from $\beta = 2.1$ to $\beta = 2.5$ and then *rises* at $\beta = 2.7$, where continuum scaling demands it keep falling. With $L/\xi_s \approx 12$ this cannot be a finite-volume squeeze. The natural reading is excited-state contamination at fixed Δ: the mass is quoted at $\Delta = 2$, and since a_t shrinks as β grows, that same Δ is a shorter *physical* Euclidean time in which to filter excited states, biasing m_{eff} upward. A second systematic points the same way — the conversion $m\,a_s = \xi\,m\,a_t$ uses the *bare* anisotropy, while the renormalized one differs and drifts with β. Both mean the $\beta = 2.7$ entry should be read as an upper bound on the mass, hence a *lower* bound on ξ_s: the true window may be somewhat wider than the tabulated factor 1.9. It would have to be very much wider — ξ_s clearing several spacings — to change the conclusion, but the honest statement is that the high-β end is bounded rather than measured.

The honest conclusion is that this is a statement about the lattice, not about the architecture. Reaching $\xi_s \gg 1$ requires either a larger volume at larger β (cost $\propto L^3$) or a milder anisotropy: since $\beta_s = \beta/\xi$ sets a_s, lowering ξ from 3 towards 1.5 at fixed β would double β_s and should push ξ_s comfortably past one spacing while keeping $m\,a_t$ resolvable in time. Both are ensemble campaigns rather than analyses, and neither is attempted here. What the scan does deliver, at the cost of one unattended run rather than four trainings, is a quantitative reason for the omission — and, incidentally, an independent reproduction of the anchor mass.

What this section is not. The statistics above are honest but small, and it is worth recording what taking the independent unit seriously cost. The ablation was first run on 8 configurations with no error bars, where the layer-3 head read $+0.127$ and four of the eight cells came out negative. At 32 configurations with a correlated jackknife the same head reads $+0.0787 \pm 0.0145$ and the negative cells collapse to the $\lesssim 0.003$ level: the point estimate moved by more than 3σ of its own final error. A Rayleigh ratio on a handful of configurations is simply not a measurement — the same caution applies to the $m\,a_t \approx 0.43$ quoted above, whose

8-configuration value on the same chain was 0.375. The Spearman correction was milder than the ablation's: configuration-level errors come out only $1.5\times$ the naive slice-level ones, so the strong correlations (-0.56, -0.33) were never in doubt, but the quoted precision now means what it says.

What remains open is the receptive-field question, and no amount of reanalysis can settle it: deciding whether the absence of depth-wise coarsening is a property of the attention mechanism or merely of $R = 2$ requires *retraining* at larger R on the full 2000-configuration ensemble, which is GPU work. Until then the negative result stands only for the network as trained. Note that the β-scan above makes the same point from the physics side: with $\xi_s \lesssim 1$, a larger R would give the attention more room without giving it more to see.

11 Conclusions, caveats, outlook

The classical chain — Wilson action, heat-bath/overrelaxation ensemble, smeared operators, multi-level GEVP, anisotropic lattice, jackknife — resolves the $SU(2)$ 0^{++} ground state at $m\,a_t \approx 0.33$ on a $12^3 \times 24$, $\beta = 2.4$, $\xi = 3$ ensemble, after the isotropic lattice demonstrably could not. On top of it, a gauge-equivariant network trained on the Rayleigh-quotient loss, constrained to a single timeslice so the transfer-matrix bound applies, and fed the same smearing levels as the classical basis, saturates the variational bound at the anchor mass and beats the classical GEVP where it can legitimately be beaten: 3.9σ less excited-state contamination at $\Delta = 1$, identical mass on the plateau, and a single learned operator that variationally subsumes the four-operator hand-built basis. In the field's standard quoting (Section 9.3): fitted masses 0.332(27) vs 0.340(30) — the same physics — with ground-state overlap $A_0 = 0.90(5)$ against 0.84(6) for the optimal classical combination, a correlated difference of $+0.066 \pm 0.031$; the projected GEVP gains only ~ 0.03 of overlap over its best single smearing level, the learned operator ~ 0.10 over the whole basis. A from-scratch replication on an independently sampled ensemble (Section 9.4) reproduces all of the operator-quality statements — $\Delta A_0 = +0.089 \pm 0.030$, a 4.5σ correlated m_{eff} difference at $\Delta = 1$ — and the two ensembles combine to $\Delta A_0 = +0.078 \pm 0.022$ (3.6σ).

Honest boundaries of the claim. (i) Two independent ensembles now, but still one (β, ξ) and coarse a_s — no continuum statement is made, for the learned operator exactly as for the classical one (Section 7.4); and the replication shows the mass itself fluctuates by $\sim 1\sigma$ between ensembles, so the "anchor" carries its own ensemble error. (ii) The 3.9σ and the overlap differences are statements about *operator quality*, not about the mass value, on which the two methods agree — as the variational principle says they must. (iii) Still open: a matched-parameter L-CNN baseline on the identical per-timeslice task. (iv) The attention readout of Section 10 rests on 32 configurations, enough to resolve its one large intervention effect at 5.4σ but not the sub-0.01 structure around it; and it reveals that the learned operator is only approximately a cubic-group scalar. (v) Whether the attention range tracks the correlation length is *not* measurable on this lattice family: the classical β-scan of Table 6 puts ξ_s below 1.1 spatial lattice spacings everywhere it can be measured, which is smaller than the grid the attention lives on.

Beyond validation, the trained operator opens a door the classical one cannot: its gauge-invariant attention maps are a direct image of the spatial loop structure the *optimal* glueball operator attends to — what the learned 18% beyond the smearing ladder actually computes is now an interpretability question with a physical answer. A first readout (Section 10) already returns four concrete findings: the operator's advantage is carried almost entirely by a single *local* attention head (5.4σ under ablation, against nothing else above 2σ); the hypothesized RG-like widening of the receptive field with depth does *not* occur at $R = 2$, several heads having collapsed to fixed maximal-range kernels; the most distributed head measurably extends its reach on high-action regions and contracts on smooth vacuum; and those frozen heads are locked to single spatial axes, so the learned operator is a cubic-group scalar only approximately. The last of these is the most actionable: an explicit A_1^{++} projection — averaging over the 24 lattice rotations — is cheap, physically motivated, and untested.

References

[1] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10** (1974) 2445.

[2] B. Berg, *Plaquette-plaquette correlations in the SU(2) lattice gauge theory*, Phys. Lett. B **97** (1980) 401.

[3] K. Osterwalder and E. Seiler, *Gauge field theories on a lattice*, Ann. Phys. **110** (1978) 440.

[4] M. Creutz, *Monte Carlo study of quantized SU(2) gauge theory*, Phys. Rev. D **21** (1980) 2308.

[5] A. D. Kennedy and B. J. Pendleton, *Improved heatbath method for Monte Carlo calculations in lattice gauge theories*, Phys. Lett. B **156** (1985) 393.

[6] M. Creutz, *Overrelaxation and Monte Carlo simulation*, Phys. Rev. D **36** (1987) 515.

[7] M. Albanese et al. [APE Collaboration], *Glueball masses and string tension in lattice QCD*, Phys. Lett. B **192** (1987) 163.

[8] G. P. Lepage, *The analysis of algorithms for lattice field theory*, in *From Actions to Answers* (TASI 1989), World Scientific (1990).

[9] C. Michael and I. Teasdale, *Extracting glueball masses from lattice QCD*, Nucl. Phys. B **215** (1983) 433.

[10] M. Lüscher and U. Wolff, *How to calculate the elastic scattering matrix in two-dimensional quantum field theories by numerical simulation*, Nucl. Phys. B **339** (1990) 222.

[11] B. Blossier, M. Della Morte, G. von Hippel, T. Mendes and R. Sommer, *On the generalized eigenvalue method for energies and matrix elements in lattice field theory*, JHEP **04** (2009) 094, arXiv:0902.1265.

[12] C. J. Morningstar and M. Peardon, *The glueball spectrum from an anisotropic lattice study*, Phys. Rev. D **60** (1999) 034509, arXiv:hep-lat/9901004.

[13] N. Madras and A. D. Sokal, *The pivot algorithm: A highly efficient Monte Carlo method for the self-avoiding walk*, J. Stat. Phys. **50** (1988) 109.

[14] M. Favoni, A. Ipp, D. I. Müller and D. Schuh, *Lattice gauge equivariant convolutional neural networks*, Phys. Rev. Lett. **128** (2022) 032003, arXiv:2012.12901.